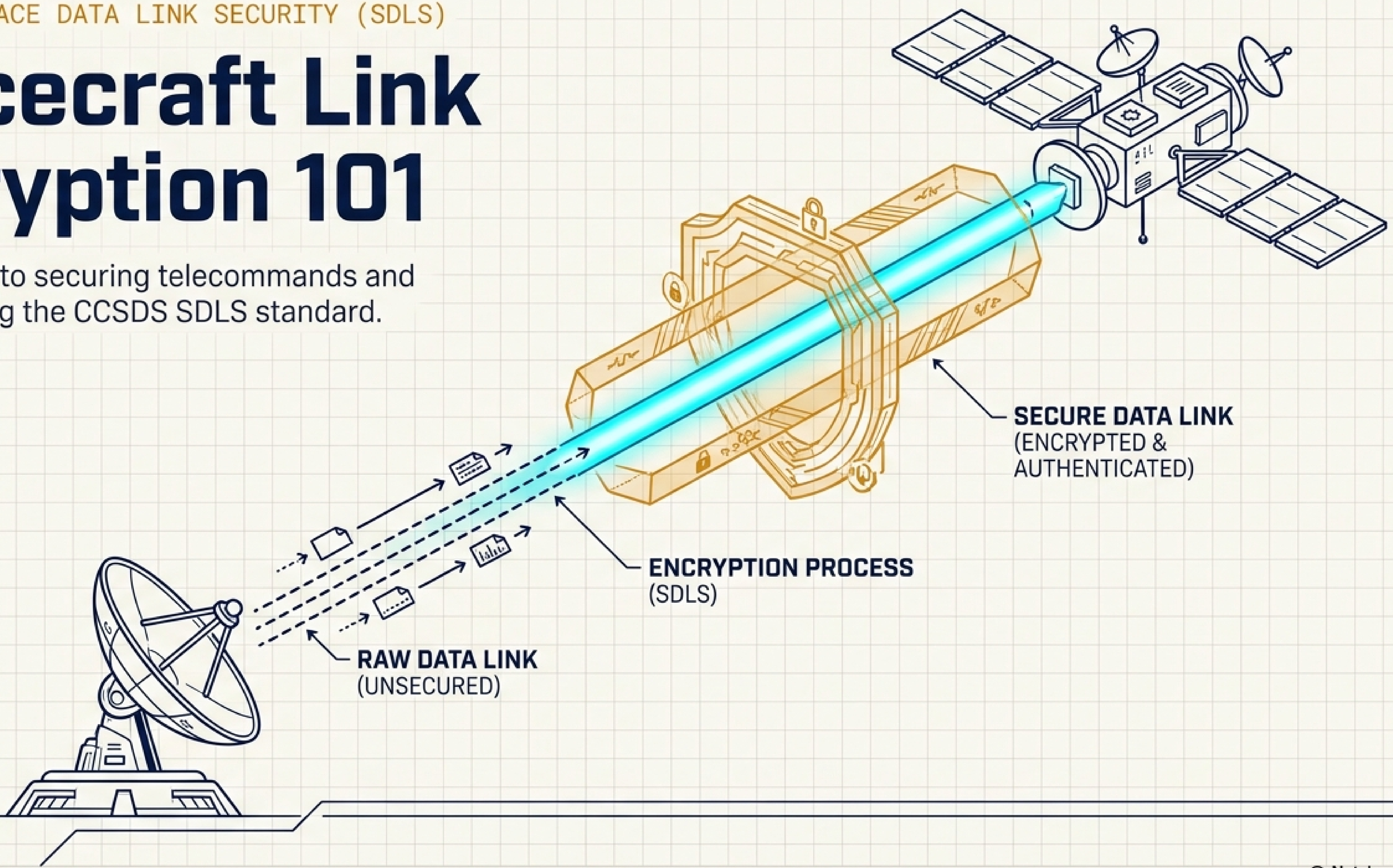
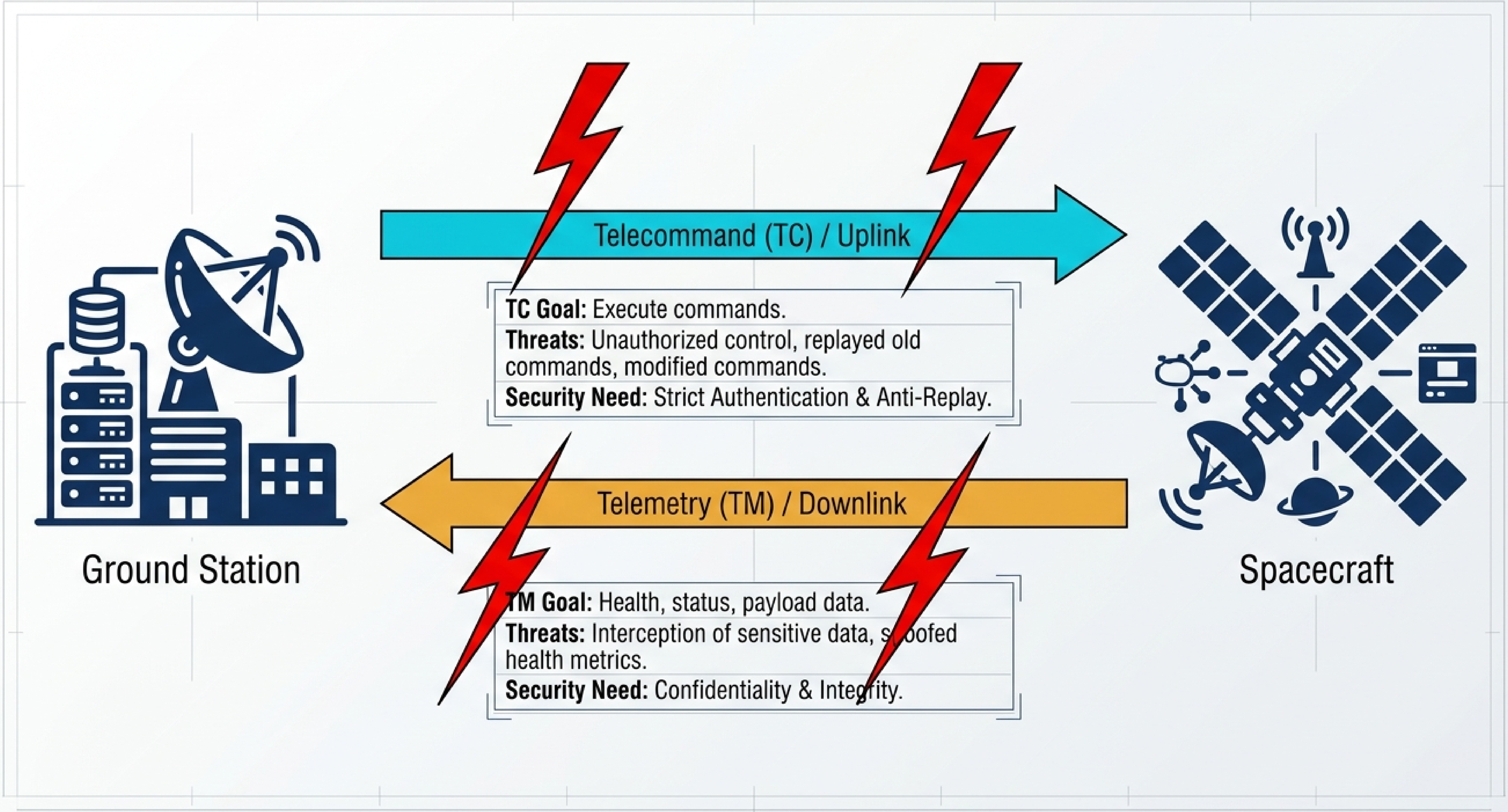
PROTOCOL: SPACE DATA LINK SECURITY (SDLS)

Spacecraft Link Encryption 101

A visual guide to securing telecommands and telemetry using the CCSDS SDLS standard.




Telecommand (TC) / Uplink
TC Goal: Execute commands.
Threats: Unauthorized control, replayed old commands, modified commands.
Security Need: Strict Authentication & Anti-Replay.
Telemetry (TM) / Downlink
TM Goal: Health, status, payload data.
Threats: Interception of sensitive data, spoofed health metrics.
Security Need: Confidentiality & Integrity.
Ground Station
Spacecraft

The Core Mental Model

[1. SPI]

Security Parameter Index: The label. Transmitted in the header, it tells the receiver which security profile to load.

[2. Security Association]

The SA: The rulebook. The database entry that defines exactly how to process the frame.

[3. Crypto Behavior]

Execution: The protocol applies the specific algorithm, key, and sequence state dictated by the SA.

Crucial Insight: The SPI is NOT the key. It is merely the index number that maps to the rulebook.

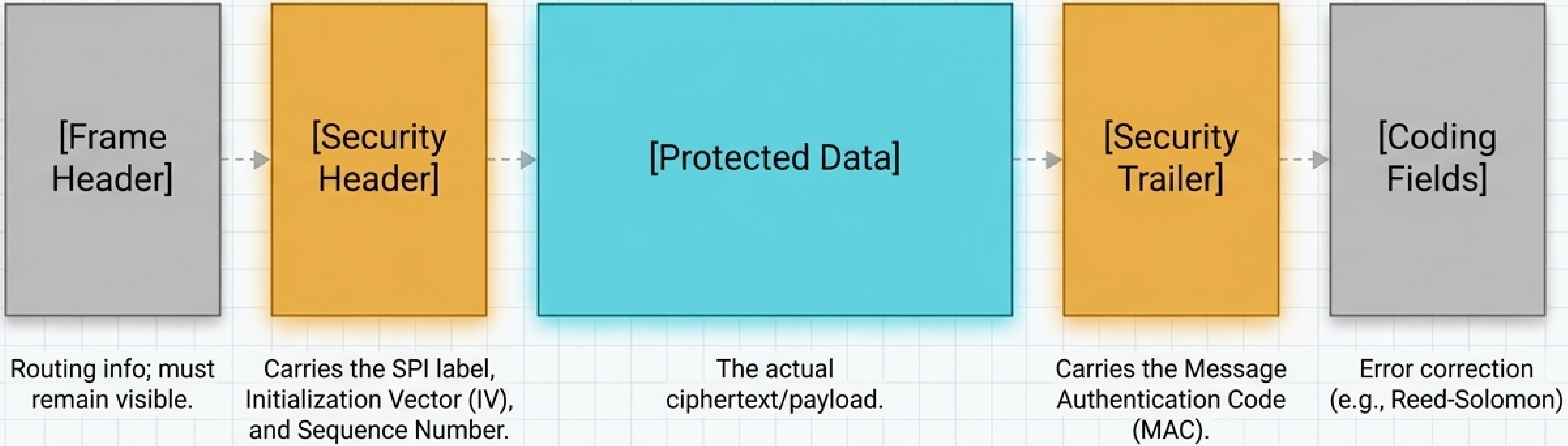
The Security Association: A One-Way Contract

SAs are **simplex** (one-way) and stateful. You do not have one SA for the mission. Uplink and downlink use completely different rulebooks.

Ground-to-Space SA (Uplink)		Space-to-Ground SA (Downlink)	
ALGORITHM	: AES-GCM	ALGORITHM	: AES-GCM
KEY MATERIAL	: [Specific Cryptographic Key]	KEY MATERIAL	: [Specific Cryptographic Key]
SEQUENCE STATE	: [Current Anti-Replay Counter]	SEQUENCE STATE	: [Current Anti-Replay Counter]
MAC LENGTH	: 128-bit	MAC LENGTH	: 128-bit

Anatomy of a Secured Transfer Frame

Key Insight: Not every part of a frame is encrypted. Link layers need visible headers to route traffic.

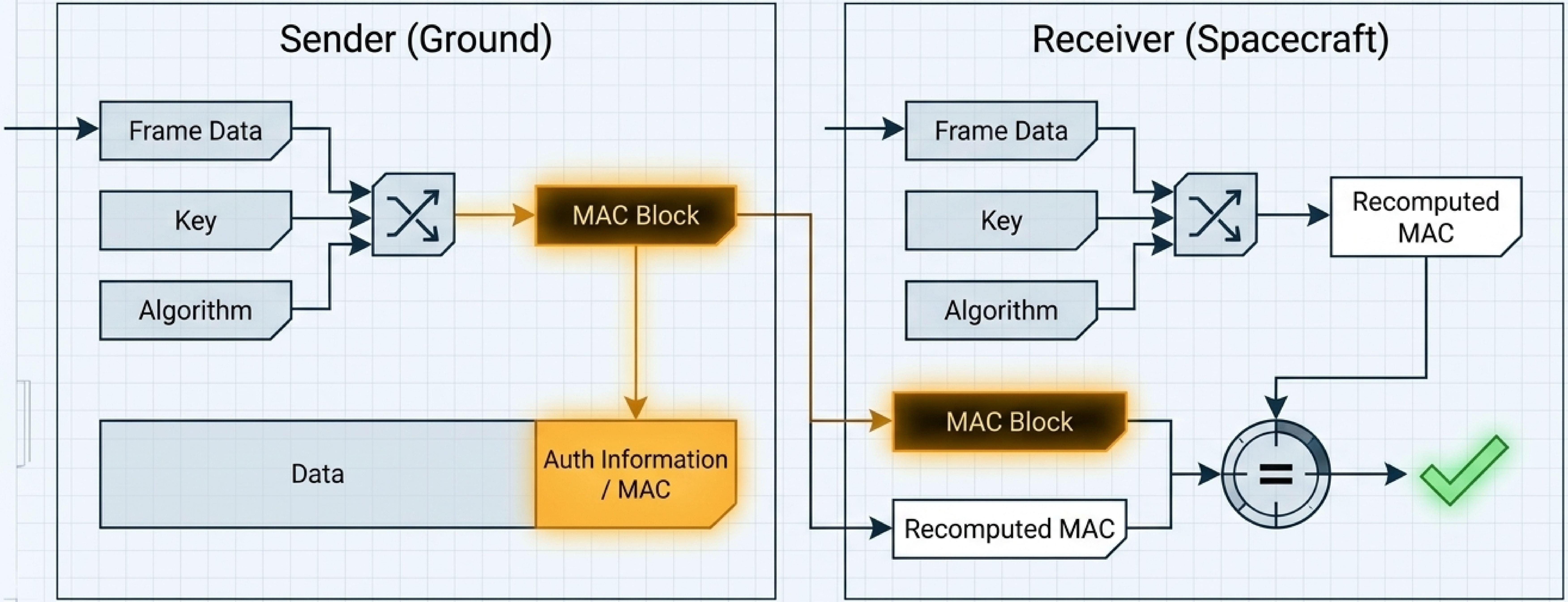


Choosing a Cryptographic Service

	Confidentiality (Hides Data)	Integrity (Detects Tampering)	Authenticity (Proves Sender)
Auth Only	✗	✓	✓
Anyone can read it, but no one can alter it.			
Encrypt Only	✓	✗	✗
Data is hidden, but an attacker might still blind-corrupt it.			
Auth + Encrypt	✓	✓	✓
The preferred modern baseline (e.g., AES-GCM).			

The MAC: A Tamper-Evident Wax Seal

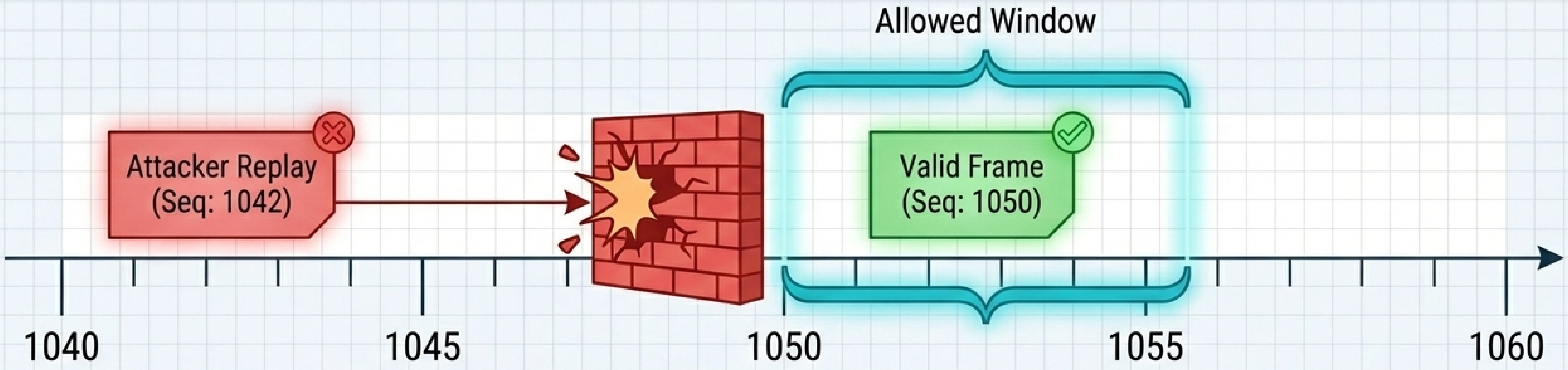
The Message Authentication Code (MAC) is the ultimate proof of integrity. If the receiver's recomputed MAC does not perfectly match the received MAC, the frame is instantly rejected.



Sequence Numbers & Anti-Replay

Threat: Encryption isn't enough—attackers can record an encrypted command and play it back later.

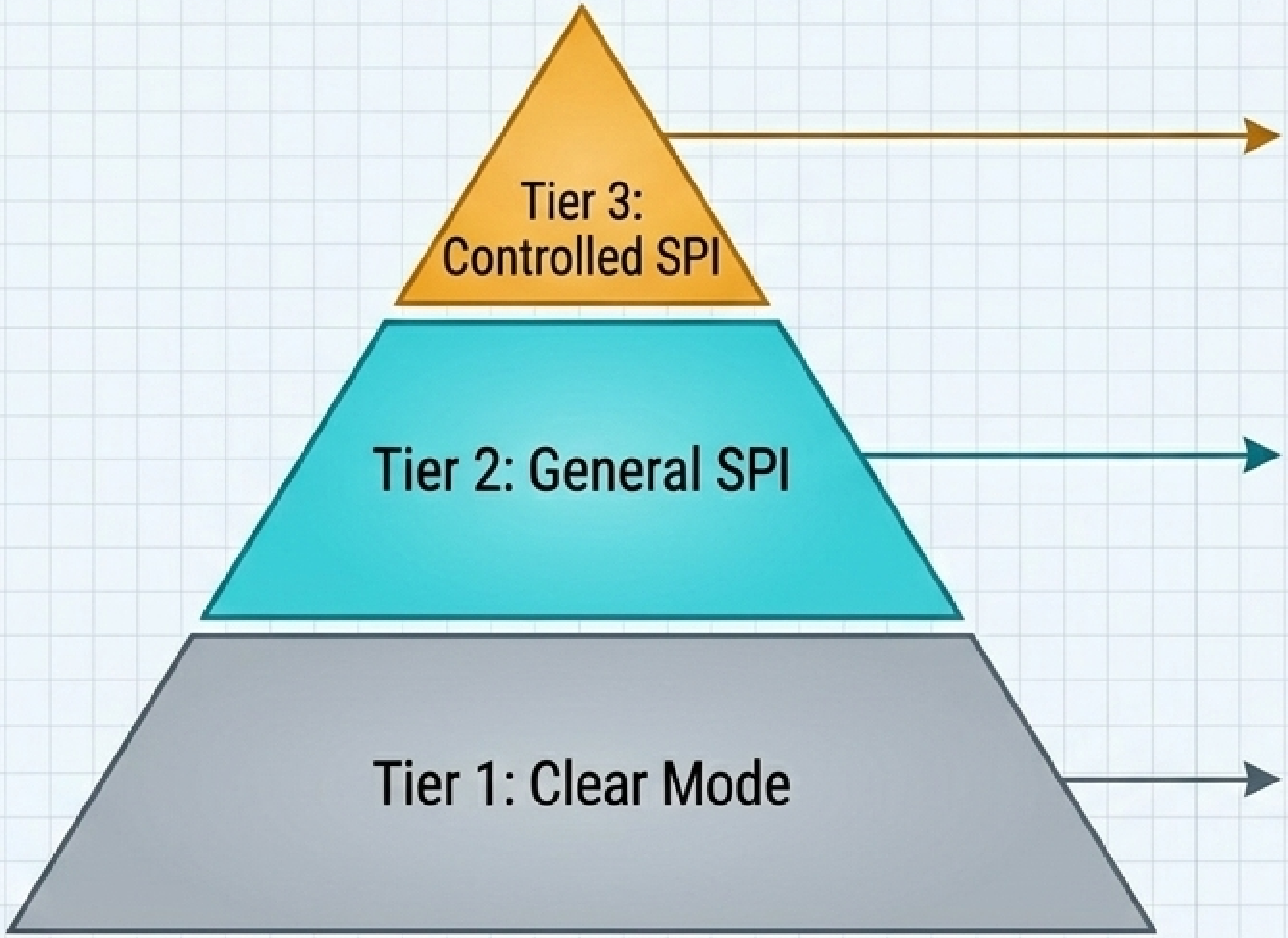
Defense: The Security Header includes an anti-replay sequence number. The receiver verifies it falls within an expected allowed window.



Critical Operational Warning: If the ground and spacecraft lose sequence synchronization, perfectly valid commands will start failing security verification and will be dropped.

The SPI Hierarchy: Command Authority

Selecting the wrong SPI is not just a formatting error; it dictates how privileged a command is.



Restricted, high-authority SA. Reserved for critical commanding, propulsion enablement, or key management.

Used for normal, low-risk operational traffic (e.g., routine housekeeping telemetry).

Unencrypted. High risk. Used strictly for safe-mode recovery, early launch acquisition, or debugging. Must be heavily monitored.

The Decryption Gate: Receiver-Side Processing

1. Parse visible header -> Read SPI.

[REJECT & LOG]

Fails if format is corrupt

2. Lookup SA database.

[REJECT & LOG]

Fails if SA is inactive or unprovisioned

3. Check Sequence Number.

[REJECT & LOG]

Fails if out-of-sync or replayed

4. Verify MAC.

[REJECT & LOG]

Fails if data altered or wrong key used

5. Decrypt Payload & Execute.

Frame Security Reports (FSR) & Troubleshooting

When a frame hits a Reject branch, the spacecraft sends a return receipt via telemetry (in the Operational Control Field / OCF).

`[ALARM: Invalid MAC]` → **Cause:** Wrong key, wrong algorithm config, or in-transit data corruption.

`[ALARM: Invalid Sequence]` → **Cause:** Replay attack, ground/spacecraft counter loss of sync, or counter reset.

`[ALARM: Invalid SA]` → **Cause:** SPI used is not provisioned, is currently inactive, or mapped to the wrong channel.

Key Lifecycle and Management

SDLS Extended Procedures allow remote management. Missions rely on a two-tier architecture to limit cryptographic exposure.

Master Keys:

- Heavily guarded, rarely used. Their sole purpose is to securely distribute and unwrap new Session Keys.

Session Keys:

- Used for daily traffic protection.
- Rotated frequently to limit exposure to cryptanalysis.

SDLS Operational Architecture at a Glance

SPI
The index label.
(Points to the SA).

SA
The rulebook.
(Defines the parameters).

Key
The secret material.
(Powers the crypto).

Secured Transfer Frame

Seq Number
Freshness & Anti-replay.
(Protects the timeline).

MAC
The tamper-proof seal.
(Proves integrity).

FSR
The return receipt.
(Reports security health).

Keys are the secret material. SAs define how keys are used. SPIs identify which SA a frame uses.